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Modular Multi-Rotary Joints SPS Concept — Challenges and Design Considerations

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***Abstract*— As a major space-based energy infrastructure in respond to the challenge of climate change, Space Solar Power (SSP) has attracted extensive international attention for more than 55 years. Recently, various SSP concepts have been proposed, aiming to efficiently harvest Giga-Watts of solar power over Earth's geostationary orbit (GEO) and transmit it to Earth wirelessly. Based on the Multi-Rotary joints Solar Power Satellite (MR-SPS) concept proposed in 2014, this paper presents an updated Modular MR-SPS (MMR-SPS) concept, together with the challenges and design considerations. The primary objective is to address the challenges associated with long-distance DC electric power transmission and exceptionally high-precision microwave power transmission (MPT). The MMR-SPS employs multiple modules, each further including a solar sub-panel and an antenna sub-panel. The solar sub-panel and antenna sub-panel in each module are connected with each other via conductive rotary joints with a much shorter distance compared with that in the MR-SPS concept. As a result, lower voltage in electric power transmission is allowed, which significantly reduces the complexity of the system and increases its reliability. Retro-reflective beamforming is employed to ensure the accurate steering of the microwave power beam to the ground's receiving aperture. For the first time, this paper analyzes and validates the performance of using retro-reflective beamforming to compensate for antenna position deviations through a numerical calculation model that includes near-field effects. Additionally, the paper outlines the transportation and assembly methods, along with the considerations of the full life-cycle cost for an MMR-SPS.**

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I. INTRODUCTION

The concept of Space Power Satellite (SPS), also known as Space Solar Power (SSP), was proposed in 1968 [1]. It aims to harvest Giga-Watts of solar power by satellites over Earth's geostationary orbit (GEO) and then deliver the power to Earth wirelessly. Unlike terrestrial solar power, SSP does not suffer from nighttime limitations and weather disruptions, offering a new source of sustainable clean energy with continuous availability. Hence SPS has been considered as one of the most promising energy projects addressing the challenges of energy crisis and climate change. Various development plans and research activities related to SPS are underway in the United States, Japan, China, South Korea, and several European countries [2-8].

The basic concept of SPS is illustrated in Fig. 1. A space power satellite typically includes a solar panel and an antenna panel. The solar power is harvested and converted to electric power in direct current (DC) by the solar cells on the solar panel. The DC electric power is delivered to the antenna panel via electric cables. The hardware integrated within the antenna panel converts the DC electric power to microwave power and controls the beamforming of the

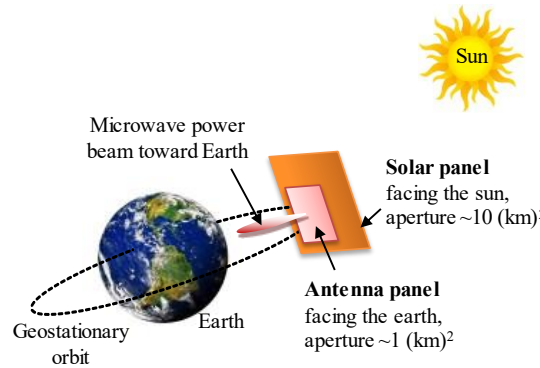


Fig. 1. Illustration of the basic concept of a Space Power Satellite (not to scale).

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Fig. 2. Illustrations of Typical SPS concepts.

microwave power toward Earth. Given the dependency of atmospheric transparency on frequencies, 5.8 GHz is considered an excellent candidate for the wireless power transmission [9]. This is also within the frequency range recommended for wireless power transmission via radio frequency beam by the International Telecommunication Union Radiocommunication sector [10]. To ensure the efficient operation of the SPS, it is preferable to keep the solar panel facing the sun and the antenna panel facing Earth. Specifically, with the SPS located over the GEO, the antenna panel facing Earth can be kept stationary while the solar panel must be dynamically adjusted toward the sun. Therefore, managing the mutual position relationship between the solar panel and the transmitting antenna panel has become one of the core issues in the SPS design. Another technical issue in SPS design is minimizing the power loss during electric power transfer from the solar panel to the antenna panel. Space environmental factors prohibit the use of high voltage techniques commonly employed in terrestrial electric power systems, resulting in significant power dissipation during long-distance power transmission [11] [12]. Dozens of SPS concepts have been proposed to tackle these challenges.

Fig. 2 illustrates some of the recently proposed concepts (comprehensive review of all SPS concepts is beyond this paper's scope). The MR-SPS [13] and the K-SSPS [7] concepts use conductive rotary joints to maintain Sun-pointing of the solar panel and Earth-pointing of the antenna panel. These designs enable a consistent and uninterrupted generation of power, but requires high-power conductive rotary joints and long cables to deliver electric power from the solar panel to antennas. The SPS-ALPHA

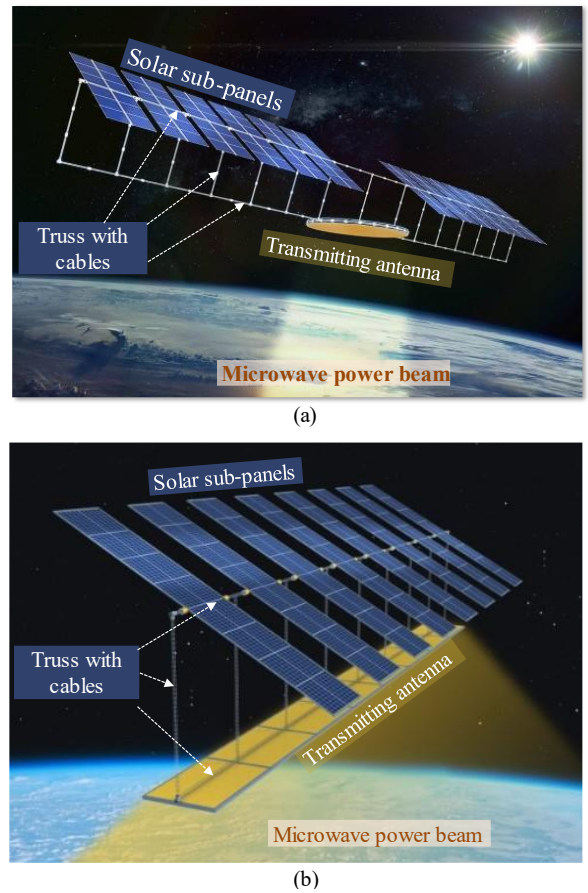


Fig. 3. The architecture of (a) the MR-SPS concept compared with (b) the proposed MMR-SPS concept.

[14], SSPS-OMEGA [15] and CASSIOPeiA SPS concepts [16] utilize specialized concentrator to redirect sunlight toward the solar panel, eliminating the need for high-power conductive rotary joints and lengthy electric cables. However, challenges arise in achieving precise control and effective heat dissipation for the concentrator. The Tethered-SPS [17] and the Caltech SPS Concepts [18][19] feature a fixed position of the solar panel relative to the antenna, resulting in varying angles between the solar panel and the sun during orbit, which causes significant fluctuations in collected solar energy during the orbital period. Effectively managing these fluctuations constitutes a main challenge in these configurations.

Based on the Multi-Rotary joints Solar Power Satellite (MR-SPS) concept proposed in 2014, this paper presents an updated Modular MR-SPS (MMR-SPS) concept, which aims to address the challenges associated with long-distance DC electric power transmission [20]. As is illustrated in Fig. 3(a), the solar panel of the MR-SPS is divided into multiple sub-panels. Multiple rotary joints along the upper truss structure are employed to orient the solar sub-panels toward the sun, which significantly reduces the power handling requirement for each conductive rotary joint. The issue of single-point of failure

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in the conductive rotary joint is also avoided. However, the difficulty of long-distance DC electric power transmission remains a substantial technical challenge. This is because the antenna panel is in the form of a circular disk aggregated at the center of the satellite, which necessitates the transmission of electric power from the solar sub-panels via cables spanning several kilometers. Due to the limitation on the voltage level that can be used in space, the power loss over such distance is significantly high. To address this challenge, the MMR-SPS employs a linear rectangular antenna panel as illustrated in Fig. 3(b). The rectangular antenna panel comprises multiple sub-panels distributed along the lower truss structure beneath the solar sub-panels. Each solar sub-panel combines with one antenna sub-panel to form an independent solar power generation and microwave power transmission (SPG-MPT) module. The power generated by each solar sub-panel is transmitted to the corresponding antenna sub-panel through much shorter cables. This design significantly mitigates the challenges associated with space power transmission and management, while also simplifying the SPS transportation and assembly.

Another technical challenge addressed in this paper is the demand for extremely high accuracy in beamforming and steering in microwave power transmission (MPT) from the antenna panel on the satellite to the receiving aperture on Earth. The modular configuration of the MMR-SPS leads to a unique long-strip-shaped antenna array layout, presenting a distinct challenge in beamforming control design. Particularly concerned in this paper is the effect of antenna position deviations and its compensation via retro-reflective beamforming.

In the subsequent sections of this paper, the system architecture of the proposed MMR-SPS concept is explored in Section II. The technique of achieving exceptionally high-precision MPT based on retro-reflective beamforming under the influence of antenna position deviations is deliberated in Section III. Section IV presents a proposed in-orbit transportation and assembly scheme, followed by an analysis of the life-cycle cost in Section V. The conclusion of this research is drawn in Section VI.

II. SYSTEM ARCHITECTURE OF THE PROPOSED MMR-SPS

As an extremely complex engineering system, an SPS architecture includes a large number of technical elements, and it is impossible for this section to cover all of them. Rather, three major elements of the MMR-SPS concept, including the SPG-MPT module design, electric power transmission and management, the microwave power transmission, are portrayed in the following subsections. Several important technical elements not covered by this paper (such as structural framework, attitude and orbit control, thermal management, information and system operation management) will be investigated separately in other papers.

The following discussions are based on the predictive energy efficiency chain of a 1 GW MMR-SPS shown in Table I. The table outlines the target energy efficiencies due to various factors considered in the proposed MMR-SPS concept. Some of them are estimated from the published researches. For instance, the best research-cell efficiency chart in [21] shows that certain photovoltaic technologies, such as multijunction cells, have achieved efficiencies higher than 40%. Values of other efficiencies are based on the estimations provided in [22][23], which are deemed to be reasonable targets given the expected progress in the related fields over the coming decades. Some of the factors are unique to the MMR-SPS concept, including the sun-pointing error, angle of sunlight and space environment effects [24]. The overall system efficiency is the cumulative energy efficiency in Table I, 12.3%. In order to generate 1 GW DC electrical power output on Earth, the solar power incident upon the solar panel of the SPS should reach around 8.2 GW. Given the solar constant to be 1.367 kW/m^2 [25], the area of the solar panel should be at least (8.2 GW) / (1.367 kW/m^2) = 6×10^6 m^2. This leads to the physical dimensions of the solar panel adopted in the MMR-SPS as 600 m × 10,000 m.

TABLE I
Predictive energy efficiency chain

Factor	Factor Efficiency	Cumulative Efficiency
Efficiency in solar power collection and conversion: 0.29		
Solar Cell	0.40	0.4
Error of Sun-Pointing	0.99	0.396
Solar Array Filling Factor	0.85	0.336
Angle of Sunlight	0.958	0.322
Space Environment Effects	0.90	0.290
Efficiency in DC power transmission and management: 0.89		
DC-DC in solar array	0.97	0.281
Rotary joints and cables	0.95	0.267
DC-DC in Antennas	0.97	0.259
Service Devices Consumption	0.999	0.258
Efficiency in DC-microwave conversion and radiation: 0.76		
DC-Microwave	0.80	0.207
Microwave radiation	0.95	0.197
Efficiency in microwave transmission in atmosphere: 0.95		
Microwave transmission	0.95	0.187
Efficiency in microwave power collection and conversion: 0.69		
Beam collection efficiency	0.9	0.168
Receiving antenna efficiency	0.9	0.151
Microwave-DC	0.85	0.129
Efficiency in DC power combination and conversion: 0.96		
Power Collection	0.98	0.126
DC-DC	0.98	0.123

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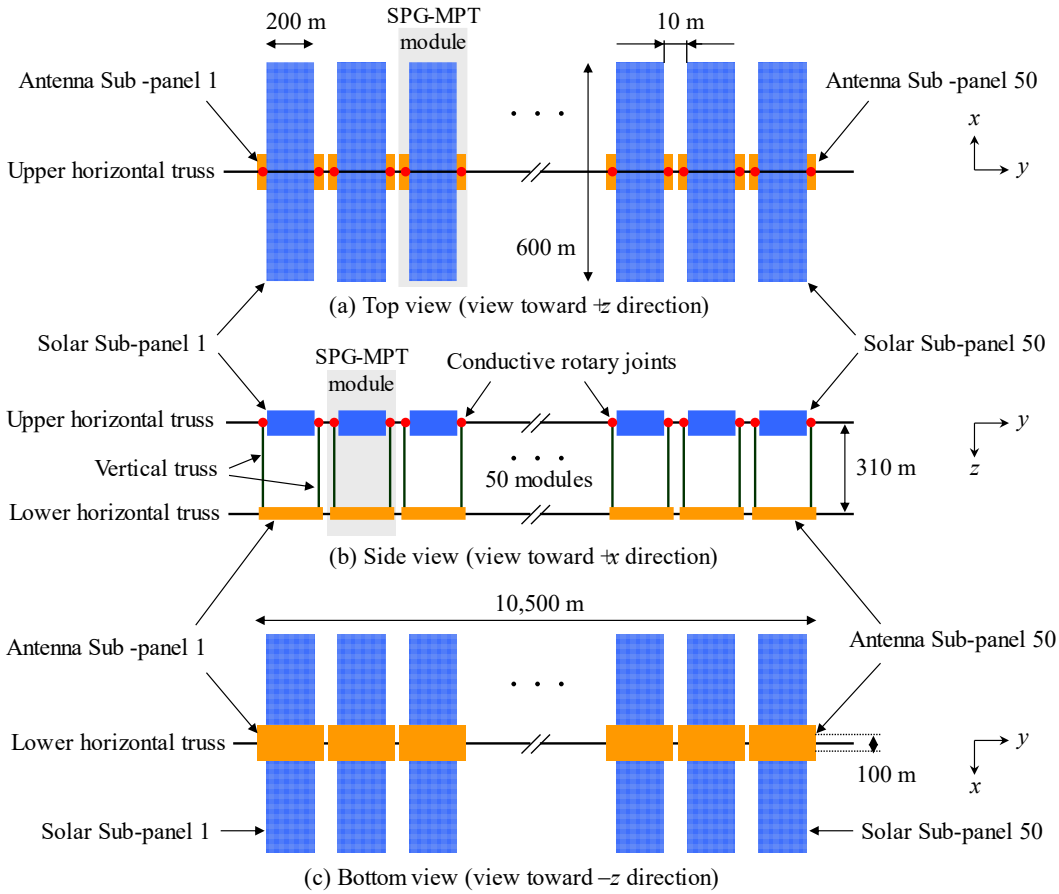


Fig. 4. Modular architecture of the solar panel and the antenna panel of the MMR-SPS concept.

A. The SPG-MPT module design

Fig. 4 depicts the specific configuration of a typical 1 GW MMR-SPS architecture. It comprises 50 SPG-MPT modules, resulting in a total length of approximately 10,500 m. Note that this estimated length is slightly larger than required as gaps among sub-panels and modules need to be taken into consideration. The upper horizontal truss supports 50 solar sub-panels. Each solar sub-panel measures 200 m × 600 m and rotates around the truss via two conductive rotary joints to face the sun accurately. The lower horizontal truss accommodates 50 antenna sub-panels, each measuring 210 m × 100 m. The upper and lower south-north trusses are interconnected by 100 vertical trusses 310 m in length. Platform service devices are installed on the solar sub-panels, antenna sub-panels, and the truss structures.

The specific physical configuration of the 50 solar sub-panels in the proposed design is illustrated in Fig. 5. The 50 solar sub-panels are arranged in the south-north direction (y direction). Each solar sub-panel includes 12 solar array modules. Each solar array module, measuring 100 m × 100 m and weighing about 3 tonnes, is manufactured and folded on Earth, launched to the orbit and then expanded to

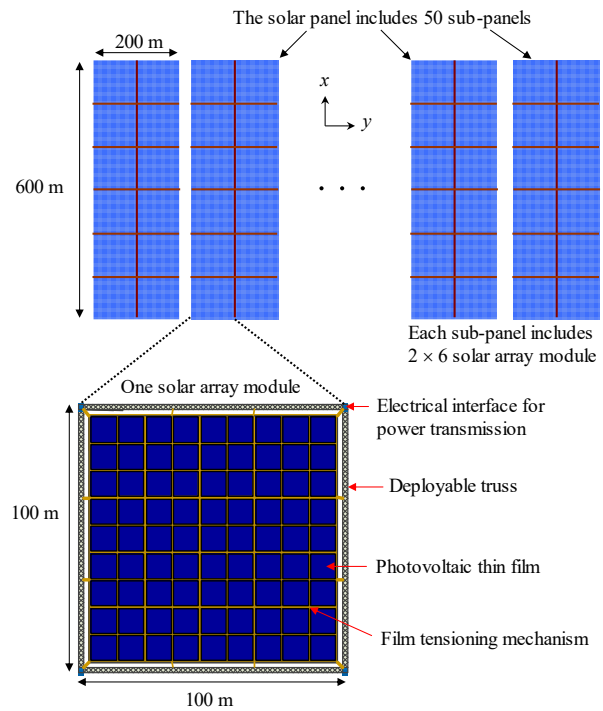


Fig. 5. Specific configuration of the solar panel.

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the full dimension before being assembled. The deployable truss and the film tensioning mechanism ensure that the photovoltaic film is flat after deployment. The power transmission interface handles the power output of the solar array module, and the assembly mechanism facilitates structural connections both among the solar array modules and between the solar array module and the structure truss [24]. The thin photovoltaic film is made of GaAs cells. The conversion efficiency from solar power to electric power is expected to reach 40%. With the sun-pointing error, solar array filling factor, angle of sunlight and space environment effects taken into consideration, the cumulative efficiency in solar power collection and conversion is estimated to be 29% as indicated in Table I. With 1.367 kW/m^2 as the incident solar power density, the electric power output of each solar array module is about (1.367 kW/m^2) × (100 m × 100 m) × 29% = 4.0 MW. The total electric power produced by one solar sub-panel is (4.0 MW per array module) × (12 array modules per sub-panel) = 48 MW. The total amount of electric power produced by the entire solar panel is (48 MW per sub-panel) × (50 sub-panels) = 2.4 GW, approximately 29% of the 8.2 GW incident solar power. The gross weight of the solar panel is approximately 3 tonnes × 12 × 50 = 1800 tonnes.

Fig. 6 shows the physical configuration of the 50 antenna sub-panels. Each antenna sub-panel further includes 10 array modules, with support provided by three 100 m trusses and two 210 m trusses. Each antenna array module measures 100 m × 20 m and weighs approximately 8.4 tonnes (with an estimated areal density of 4 kg/m²). The total weight of the antenna panel is about (8.4 tonnes per array module) × (10 array module per sub-panel) × (50 sub-panels) = 4200 tonnes. Like the solar array modules, the antenna array modules are manufactured on Earth, physically folded, and then launched to the orbit for assembly. The antenna panel does not have to be completely planar after deployment, as a technique called retro-reflective beamforming can be employed to compensate the physical deformation of antenna panel,

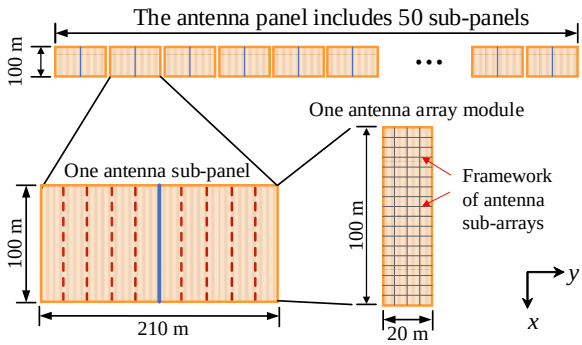


Fig. 6. Specific configuration of the antenna panel.

which is discussed in Section III. The 48 MW DC electric power output from each solar sub-panel is transmitted to the corresponding antenna sub-panel with an efficiency of 89% as estimated in Table I. Therefore, each antenna sub-panel accepts (48 MW× 89%) = 42.7 MW DC power and converts it to microwave. With the DC-microwave conversion and radiation efficiency estimated to be 76%, the microwave power radiated by one antenna sub-panel is approximately (42.7 MW × 76%) = 32 MW. All the radiating elements in the 50 antenna sub-panels jointly construct a narrow microwave beam carrying a total power of about (32 MW × 50) = 1.6 GW toward Earth.

B. Electric Power transmission and management

Due to the SPG-MPT modular design, the electric power transmission distance from the solar panel to the antenna panel is reduced to 310 m along the vertical truss. As a result, it allows for the use of relatively shorter cables and lower voltage (5 kV), significantly simplifying the Power Transmission and Management (PTM) system and enhances its reliability.

The block diagram of power management in one of the 50 SPG-MPT modules is exhibited in Fig. 7. The power management scheme of the other 49 modules is identical to that in Fig. 7. The output voltages of the 12 solar array modules are 500 V. They are boosted to 5 kV before being delivered to the solar sub-panel bus in order to reduce the thermal loss of the electric cables. The voltage on each solar sub-panel bus is 5 kV with a current of 4800 A through each of the two rotary joints. Roller ring conductive rotary joints can be employed as it offers the benefits of low loss, high power handling capacity, high-speed stability, and extended longevity. It has been implemented on the International Space Station as the Solar Alpha Rotary Joint [26][27]. Nevertheless, technological breakthroughs in heat dissipation, high-voltage insulation and protection are required for roller ring conductive rotary joints to handle power levels of hundreds of kilowatts and above [12]. The 48 MW output power is transmitted to the antenna sub-panel bus via the 310-meter-long cables along the two vertical trusses. It is further converted back to 500 V by voltage step-down converting units to supply power to the 10 antenna array modules. The electric power transmission and management is also equipped with a service sub-system shared by all modules. The overall efficiency of the PTM system, including the efficiencies of DC-DC in solar array, rotary joints and cables, DC-DC in antennas and service devices consumption, is 89%.

C. Microwave power transmission

As analyzed in subsection II.A, a total microwave power of 1.6 GW is radiated by the antenna panel toward Earth. As illustrated in Table I, the microwave power transmission and conversion efficiencies include the atmospheric

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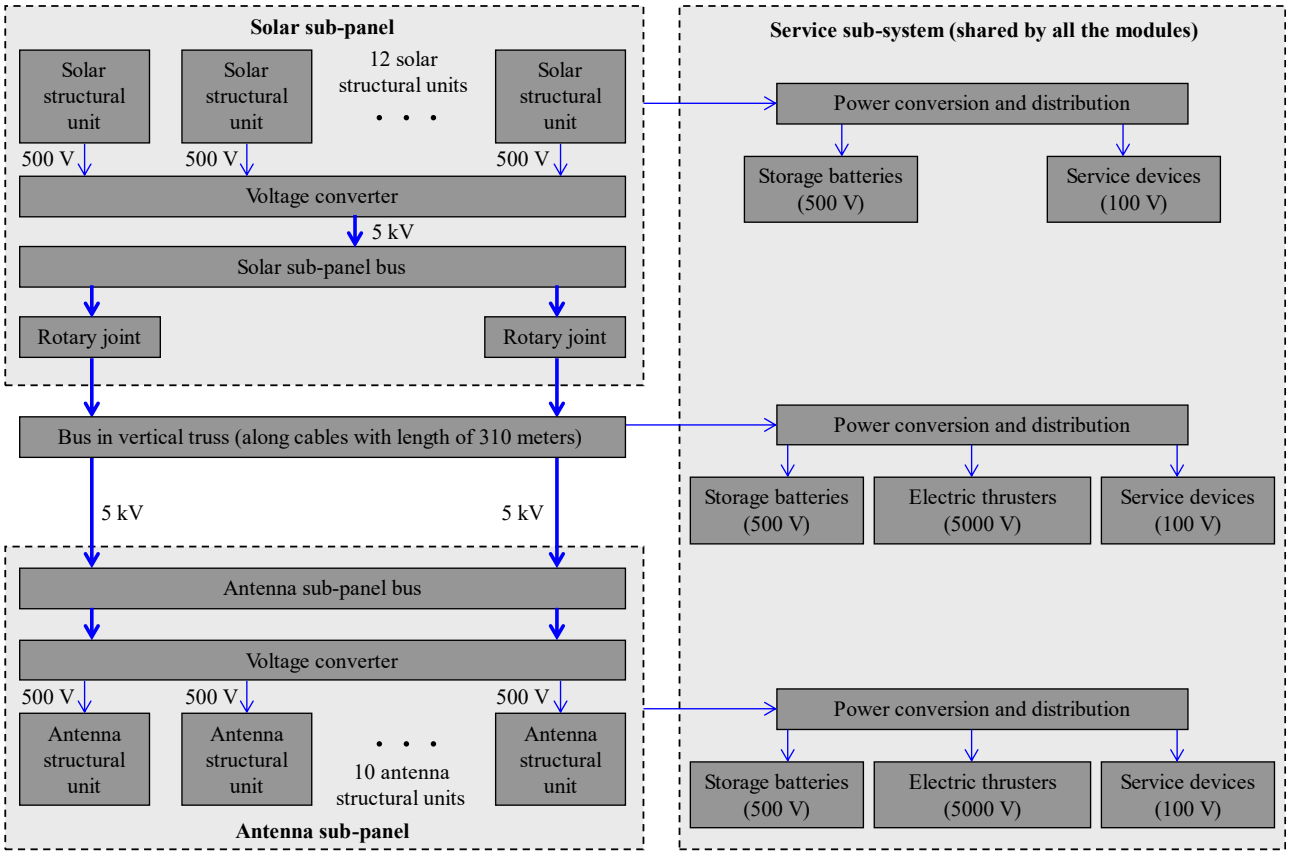


Fig. 7. Block diagram of power management in one of the 50 modules.

attenuation (estimated to be 95%), the microwave power collection and conversion efficiency (69%), and the DC power combination and conversion efficiency (96%). That leads to an overall DC output power on Earth to be (1.6 GW × 95% × 69% × 96%) =1 GW. One of the most important factors in these efficiencies is the beam collection efficiency (BCE), which is defined as the ratio of the microwave power collected by the receiving antenna on Earth to the power radiated by the antenna panel on the SPS (excluding the atmospheric attenuation). The BCE is expected to be higher than 90%. Given the microwave frequency to be 5.8 GHz, dimensions of the transmitting antenna panel to be 10,500 m × 100 m and the transmission distance to be approximately 36,000 km from GEO to Earth, a BCE of over 90% would require a receiving aperture of over 300 m × 30,000 m as estimated from the antenna theory of beamforming [28][29]. Notice that the above high BCE requirement must be achieved by near-field focusing [30]. In fact, Given the largest dimension of the solar panel $D = 10.5$ km and the wavelength $\lambda_0 = 5.17$ cm in space, the well-known far-field distance is $(2D^2/\lambda_0) = 3.9$ (million km), which is much larger than the 36,000 km transmission distance. Furthermore, the antenna array modules must be meticulously designed to mitigate their radiation along the directions of the grating lobes that arise from the inter-module gaps to ensure the achievement of the 90% BCE.

Moreover, it is necessary that the normal direction of the antenna panel accurately points toward the receiving aperture on Earth (assuming broadside radiation from the antenna panel). In practice, however, deviation from the ideal status is unavoidable. Thanks to today's sophisticated satellite control technology [31], it is highly possible that the attitude deviation of solar panel and antenna panel does not exceed 0.1°. Nevertheless, an attitude deviation in the range of [–0.1°, 0.1°] still significantly influence the BCE. An angular error of 0.1° (or 0.0017 radian) transforms to a pointing error of (36,000 km) × (0.0017 radian) = 62.8 km on Earth, which is way beyond the size of the receiving aperture. Therefore, the antenna panel needs to employ phased arrays to steer the power beam with much higher accuracy.

In a conventional SPS concept such as the MR-SPS, the typical beam steering accuracy requirement is generally considered to be 0.0005°, so that the deviation of the footprint of the microwave power beam from GEO can be constrained within 300 m on Earth. This is expected to be achieved by the transmitting antenna in the form of a circular phased array with diameter of about 1 km. However, in the proposed MMR-SPS concept, the much larger size of the transmitting antenna in the long side (10.5

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km along the y direction as shown in Fig. 4) results in a concentrated power distribution within a much narrower region along this direction. This demands a significantly higher beam steering accuracy along this specific direction. However, on a positive note, the extended 10.5 km long side of the transmitting antenna allows for incorporation of more phase-controlled antenna elements along this direction. This capability enables the implementation of a much more precise beam-control mechanism, enhancing the overall beam steering accuracy along the direction.

Referring to [32], the one-dimensional beam pointing error of a phased array antenna with beam direction control using monopulse method can be calculated by (assuming all radiating elements have the same excitation amplitude)

$$\Delta\theta = \frac{2\lambda_0}{(\pi d \cos\theta_0) M^{1.5}} \Phi \tag{1}$$

where $\Delta\theta$ is the root-mean square (rms) beam pointing error, M is the total number of linear radiating elements, Φ is the rms error of the feeding phases of the transmitting antenna, d is the spacing of the radiating elements, λ_0 is the wavelength, θ_0 is the beam pointing direction from the normal direction of the power transmitting antenna. For the MMR-SPS concept, if the regular half-wavelength element spacing is adopted (i.e., if d = 2.59 cm), a total of M = 386,000 radiating elements are distributed along the long side. Let $\cos\theta_0 = 1$, $\Phi = 20°$ (rms), the ideal beam pointing accuracy $\Delta\theta$ of the transmitting antenna along the long side direction is approximately 1.1×10^{-7} degrees (rms). However, for the half-wavelength spacing of radiating elements, there is a linear distribution of 3,860 elements along the short (100 m) side of the antenna panel, leading to a total of (386,000 × 3,860) ≈ 1.5 (billion) radiating elements. Applying independent phase control of 1.5 billion antenna elements leads to a formidably high cost. Therefore, it is practical to divide the entire antenna array into multiple sub-arrays, apply identical feed phase to all the elements within each sub-array, and apply independent phase control to the sub-arrays. Specifically, with the size of the sub-array to be 5 m × 5 m, the total number of independent phase control circuits is reduced to (2,000 × 20) = 40,000. In this case, d = 5 m and a total of M = 2000 antenna sub-arrays are distributed along the long side. With $\cos\theta_0 = 1$ and $\Phi = 20°$ (rms), the beam pointing accuracy $\Delta\theta$ given by (1) is approximately 1.5×10^{-6} ° rms, i.e., less than 10 m deviation for the 36,000 km transmission distance.

III. RETRO-REFLECTIVE BEAMFORMING OF MPT

The beam pointing accuracy estimation given by (1) only considers the random feeding phase error. Many other practical factors might also significantly affect the beam steering accuracy and eventually affect the BCE. One of the most important factors is the antenna position deviation.

While the SPS is positioned in the GEO, achieving and maintaining a completely flat and stationary kilometer-large antenna array at all times is impractical. Assembling errors can introduce deviations in the attitude and relative positions of the antenna modules, impacting the beamforming quality and pointing accuracy of the MPT system. In addition, various environmental factors also contribute to antenna position deviations and aperture deformation. For example, extreme temperature fluctuations cause expansion and contraction of the antenna structure. External forces like solar radiation pressure result in distortion and bending. Potential collisions with small meteoroids or micrometeorites lead to surface deformations and vibrations. The combined effect of these factors results in the misalignment of the antenna array aperture from its ideal planar state, detrimentally affecting beamforming performance and pointing accuracy. How to compensate for these adverse effects stands as a fundamental challenge in the MPT design for SPS. One way of achieving the compensation is by detecting the antenna position deviation with sensors and then making shape calibrations, which obviously increases the system complexity significantly. A framework to reconstruct the shape of flexible arrays via mutual coupling measurements while requiring no additional sensors is proposed in [33], which provides a promising approach for compensating the antenna position deviations effectively. This paper focuses on another technique to address this challenge, i.e., the retro-reflective beamforming [30][34-35].

In a retro-reflective beamforming scheme, MPT is accomplished via the following two steps.

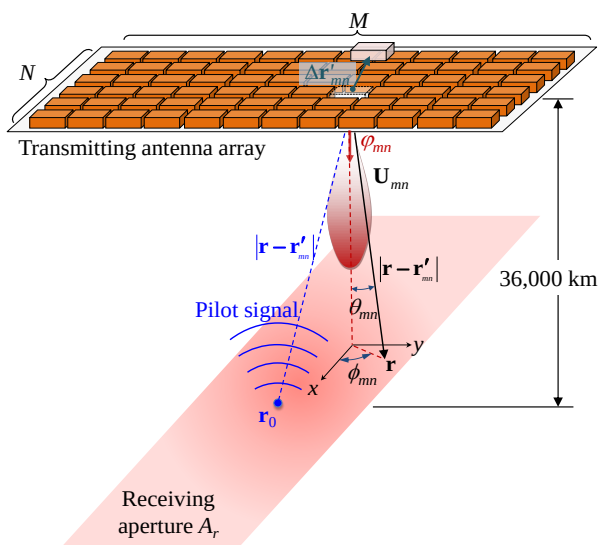


Fig. 8. Retro-reflective beamforming model of MPT in MMR-SPS (not to scale).

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Step (i): The microwave power receiver broadcasts a pilot signal as the "request for wireless power". The phase distribution of the pilot signal arriving at the antenna array of the microwave power transmitter is detected.

Step (ii): By feeding the power-transmitting array with a phase distribution conjugate to that of the received pilot signal, the microwave power transmitter generates a microwave power beam toward the power receiver.

To simulate the power density distribution over the receiving aperture and estimate the effect of the antenna position deviation on the BCE, a calculation model illustrated in Fig. 8 is used. In the model, the SPS includes a planar antenna array with $M \times N$ phased antenna elements in the x-y plane, with M rows and N columns. $\mathbf{r}'_{mn}$ is the position vector of the phase center of the antenna element in the mth row and the nth column. $\mathbf{r}$ is the position vector of an observation point on the power-receiving aperture. The phased transmitting array generates a microwave power beam focused on the location from which the pilot signal is broadcasted. This location is represented by the position vector $\mathbf{r}_0$, which is usually the center of the receiving aperture.

It is important to notice that $\mathbf{r}$ (as well as $\mathbf{r}_0$) resides in the near-field region of the planar antenna array of the SPS. However, for each single antenna element (an antenna sub-array with dimensions 5 m ×5 m) in the antenna panel, $\mathbf{r}$ is in the far-field region. Therefore, the total electric field at an observation point $\mathbf{r}$ can be written as

$$\mathbf{E}(\mathbf{r}) = \sum_{m=1}^{M}\sum_{n=1}^{N} \frac{A_{mn} e^{j\varphi_{mn}} e^{-jk_0|\mathbf{r}-\mathbf{r}'_{mn}|}}{|\mathbf{r}-\mathbf{r}'_{mn}|} \mathbf{U}_{mn}(\theta_{mn}, \phi_{mn}) \qquad (2)$$

where A_{mn} and φ_{mn} represent the magnitude and phase of the excitation of the antenna element with indices (m, n). $k_0 = 2\pi/\lambda_0$ is the wavenumber in free space. $\mathbf{U}_{mn}$ is a polarization vector representing the normalized field contribution of the antenna element with indices (m, n) to the total $\mathbf{E}$ field. $\mathbf{U}_{mn}$ depends on the direction of the observation point $\mathbf{r}$ relative to the phase center $\mathbf{r}'_{mn}$, which is a function of the angles θ_{mn} and ϕ_{mn} as illustrated in Fig. 8. Although the field contribution from each antenna element is calculated with far-field approximation, the equation (2) is different from the commonly used far-field calculation model in the following aspects.

(i) The phase delay $k_0|\mathbf{r}-\mathbf{r}'_{mn}|$ in the exponential propagation factor is evaluated rigorously, instead of assuming a linear phase variation over the antenna array aperture in the far-field approximation.

(ii) The distances $|\mathbf{r}-\mathbf{r}'_{mn}|$ in the denominator is evaluated rigorously, instead of assuming the same distance for all (m, n) in the far-field approximation.

(iii) $\mathbf{U}_{mn}$ is different for different indices (m, n) with the angles θ_{mn} and ϕ_{mn} evaluated rigorously, instead of assuming the same direction of $\mathbf{r}$ relative to $\mathbf{r}'_{mn}$ for all (m, n) in the far-field approximation.

With the above improvements in the calculation model, the near-field effect is taken into account in (2).

In the ideal case, the excitation phase φ_{mn} is determined with the retro-reflective beamforming scheme based on the principle of phase conjugation, which gives

$$\varphi_{mn} = k_0 |\mathbf{r}_0 - \mathbf{r}'_{mn}| \qquad (3)$$

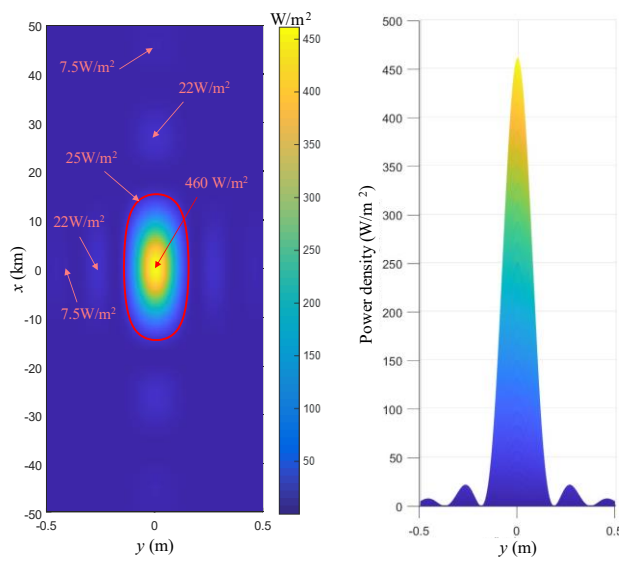
so that the microwave power beams from all the antenna elements superpose in phase on the desired observation point $\mathbf{r}_0$. In this case, the $\mathbf{E}$ field distribution over the receiving aperture can be written as

$$\mathbf{E}_{\text{ideal}}(\mathbf{r}) = \sum_{m=1}^{M}\sum_{n=1}^{N} \frac{A_{mn} e^{-jk_0(|\mathbf{r}-\mathbf{r}'_{mn}|-|\mathbf{r}_0-\mathbf{r}'_{mn}|)}}{|\mathbf{r}-\mathbf{r}'_{mn}|} \mathbf{U}_{mn} \qquad (4)$$

With the size of phase-controlled antenna elements to be 5 m × 5 m, M = 2000, and N = 20, the simulated power density distribution on the ground (Fig. 9) reveals a main lobe zone of about 30 km × 300 m with a BCE of 85% and the highest power density reaching about 460 W/m². The first side lobe zone and second side lobe zone have the highest power densities of approximately 22 W/m² and 7.5 W/m², respectively. To achieve a higher BCE, the rectenna array on the ground is designed to cover an area of 60 km × 800 m.

However, in the presence of transmitting aperture deformation and element position deviations, there will be a degradation of the beamforming performance as the $\mathbf{E}$ field distribution is disturbed as

$$\mathbf{E}_{\text{deform}}(\mathbf{r}) = \sum_{m=1}^{M}\sum_{n=1}^{N} \frac{A_{mn} e^{-jk_0(|\mathbf{r}-\mathbf{r}'_{mn}-\Delta\mathbf{r}'_{mn}|-|\mathbf{r}_0-\mathbf{r}'_{mn}|)}}{|\mathbf{r}-\mathbf{r}'_{mn}-\Delta\mathbf{r}'_{mn}|} \mathbf{U}_{mn} \qquad (5)$$



(a) (b)

Fig. 9. The power density distribution in the idea case (not to scale), (a) over the receiving aperture on the ground and (b) across the aperture with fixed x = 0.

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where $\Delta\mathbf{r}'_{mn}$ represents the position deviation of the antenna element with indices *mn* as illustrated in Fig. 8. To quantify the beamforming performance degradation, a degradation factor can be introduced as

$$\mathrm{DF}_{\mathrm{deform}} = \frac{\iint_{A_r} \left|\mathbf{E}_{\mathrm{deform}}(\mathbf{r})\right|^2 dS}{\iint_{A_r} \left|\mathbf{E}_{\mathrm{ideal}}(\mathbf{r})\right|^2 dS} \tag{6}$$

which can be considered as a decrease factor in BCE.

To compensate for the effect of deformation, retro-reflective beamforming is employed, which fixes the excitation phase as

$$\varphi_{mn} = k_0 \left|\mathbf{r}_0 - \mathbf{r}'_{mn} - \Delta\mathbf{r}'_{mn}\right| \tag{7}$$

The **E** field distribution after compensation of the retro-reflective beamforming becomes

$$\mathbf{E}_{\mathrm{retro}}(\mathbf{r}) = \sum_{m=1}^{M}\sum_{n=1}^{N} \frac{A_{mn} e^{-jk_0\left(\left|\mathbf{r}-\mathbf{r}'_{mn}-\Delta\mathbf{r}'_{mn}\right|-\left|\mathbf{r}_0-\mathbf{r}'_{mn}-\Delta\mathbf{r}'_{mn}\right|\right)}}{\left|\mathbf{r}-\mathbf{r}'_{mn}-\Delta\mathbf{r}'_{mn}\right|}\mathbf{U}_{mn} \tag{8}$$

The degradation factor after compensation, denoted by $\mathrm{DF}_{\mathrm{retro}}$, can be calculated as

$$\mathrm{DF}_{\mathrm{retro}} = \frac{\iint_{A_r} \left|\mathbf{E}_{\mathrm{retro}}(\mathbf{r})\right|^2 dS}{\iint_{A_r} \left|\mathbf{E}_{\mathrm{ideal}}(\mathbf{r})\right|^2 dS} \tag{9}$$

The antenna position deviations can be divided into two types: random position errors and overall aperture deformations. All these deviations are most likely caused by shifts of the antenna modules in the *z*-direction [36][37].

For random positional errors, two cases are considered here: uniform distributions and normal distributions of the position error in *z*-directions. In other words,

$$\Delta\mathbf{r}'_{mn} = \Delta z_{mn}\hat{\mathbf{z}} \tag{10}$$

where Δz_{mn} is a random variable with uniform or normal distributions.

For overall aperture deformations, the bent and bow deformation models proposed in [36] and the sine and saddle models proposed in [38] are adopted in the simulation, as shown in Fig. 10.

Fig. 11 shows the degradation factors due to the two types of random position errors and the bent aperture deformations along the *x* and *y* directions as functions of the *position deviation amount*. For random positional errors, the position deviation amount is defined as the standard deviation of the random variable Δz_{mn}. For the overall aperture deformations, the position deviation amount is defined as the difference between the maximum and minimum of Δz_{mn}.

From the results in Fig. 11, it is seen that without compensation of the retro-reflective beamforming, a bent deformation as small as 4 cm along the 10-km long side (*y* direction) causes a degradation factor $\mathrm{DF}_{\mathrm{deform}} = 91\%$. The same amount of bent deformation along the 100 m long side leads to $\mathrm{DF}_{\mathrm{deform}} = 88\%$. When the bent deformation amount exceeds 7 cm, $\mathrm{DF}_{\mathrm{deform}}$ below 50% is observed, indicating a substantial reduction of more than 50% in BCE. Moreover, for random antenna position errors, a tolerance of only 3 mm deviation is permissible if a degradation factor of 90% or above is to be maintained. Nevertheless, with the retro-reflective beamforming compensations, the degradation factor $\mathrm{DF}_{\mathrm{retro}}$ is maintained at almost 100% for all deviation amounts. Therefore, for the exceptionally large-scale antenna array of SPS, retro-

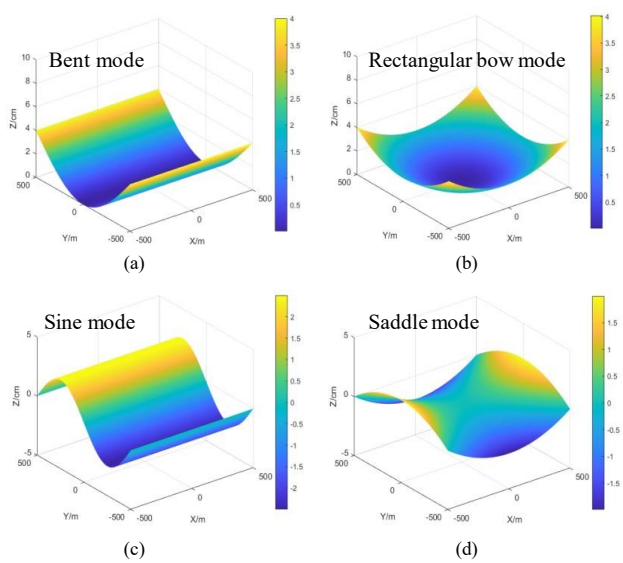


Fig. 10. Four deformation models of large-scale arrays.

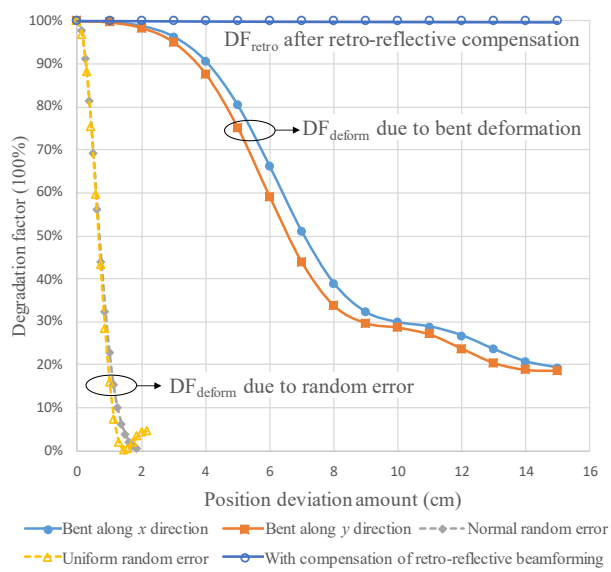


Fig. 11. Degradation factor of the 10 km × 100 m transmitting antenna array of MMR-SPS due to different types of the antenna position deviations.

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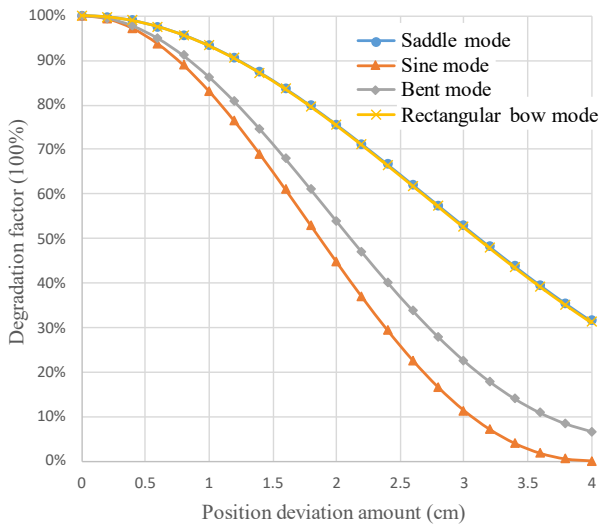


Fig. 12. Degradation factor of a 5 m × 5 m antenna element due to different types of the aperture deformations.

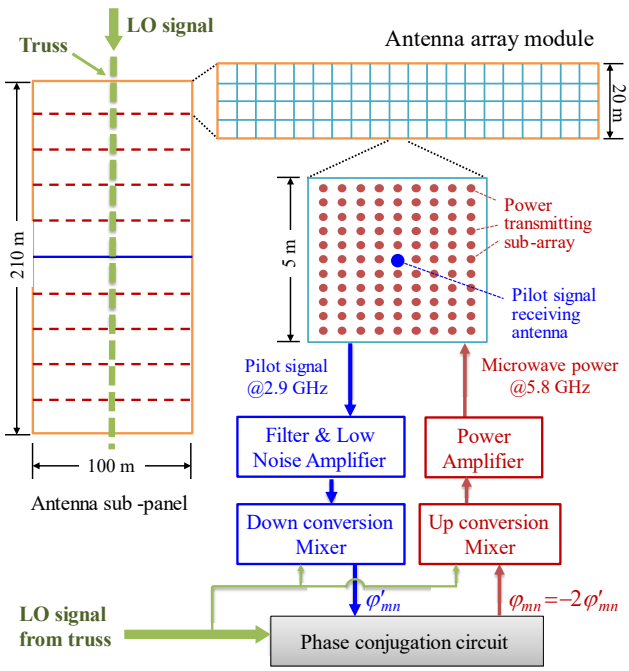


Fig. 13. Configuration scheme of the retro-reflective beamforming antennas and circuits.

reflective beamforming plays a crucial role in achieving highly efficient MPT.

It is important to note that the retro-reflective beamforming scheme cannot compensate for degradation of the beamforming performance caused by the deformation within a single-fed antenna element (sub-array). Given the antenna element size of 5 m × 5 m, it is challenging to completely avoid the inner deformation. Fig. 12 illustrates the simulated degradation factor due to four types of aperture deformations of the antenna elements. It is seen that, to maintain a degradation factor of 90% or above, the overall aperture deviation of the 5 m × 5 m antenna elements should not exceed 8 mm.

While reducing the antenna element size makes it easier to minimize the inner deformation, it comes with the price of requiring more antenna elements and circuits. This, in turn, significantly increases the system's complexity and cost.

In the above analysis, it is assumed that the pilot signal has the same frequency as the microwave power beam. In this case, a time-division duplex scheme must be used to provide isolation of the transmitted power from the received pilot signal. In practice, frequency-division duplexing can also be used. Fig. 13 illustrates a configuration scheme of the retro-reflective beamforming circuits with frequency-division duplexing. The frequency of the pilot signal is 2.9 GHz, half of the frequency of the microwave power beam. Each antenna element contains a 5 m × 5 m power transmitting sub-array and a small pilot signal receiving antenna embedded within the sub-array. For accurate retro-reflective beamforming control, the phase center of the power transmitting sub-array should overlap with the phase center of the pilot signal receiving antenna.

The received pilot signal is filtered and amplified, and then down-converted to baseband for digital processing. The digital processing is performed by a phase conjugation circuit, which converts the baseband pilot signal with phase φ'_{mn} to a conjugate baseband output with phase $\varphi_{mn}=-2\varphi'_{mn}$. The conjugate baseband signal is then up-converted to 5.8 GHz, amplified and fed to the power transmitting sub-array. The power transmitting circuit, pilot signal receiving circuit, and the phase conjugation circuit are installed on the back face of each antenna element (sub-array), all synchronized by a local oscillator (LO) signal for precise timing.

The LO signal originates from a single sinusoidal signal generator and is distributed to each antenna array module via the lower south-north truss, as is shown in Fig. 13. As one of the major technical challenges of the SPS deign, distribution of the LO signal for the retro-reflective beamforming circuits calls for a phase-stabilized frequency transfer network spanning kilometers, which is still undergoing investigation.

IV. TRANSPORTATION AND ASSEMBLY IN ORBIT

The obviously enormous mass and volume of SPS pose significant challenges for its transportation and assembly in orbit, necessitating division into multiple modules for launch into orbit based on the transport capacity of the launch vehicle. MMR-SPS adopts the GEO assembly and transportation mode. The process involves utilizing heavy-lift rockets to launch the SPS modules into Low Earth Orbit

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(LEO). Then, the upper stage of the rocket or an inter-orbit transporter is employed to transport all the SPS modules to GEO, where assembly takes place (refer to Fig. 14).

Key features of this approach include conducting all assembly processes in GEO, which significantly reduces the risk of collisions with space debris. This is due to the substantially lower concentration of debris present in GEO compared to lower orbits, thereby enhancing the safety and reliability of the assembly operations. The modular transportation strategy results in a relatively low transportation capacity requirement for the inter-orbit transporter, albeit necessitating more inter-orbit transporters for support. The impact of Earth's radiation belt on the SPS modules during orbit transfer is reduced by the protective shield of the inter-orbit transporter. Moreover, the inter-orbit transporter can achieve long-term reusability through propellant refueling and the solar array replacement in LEO.

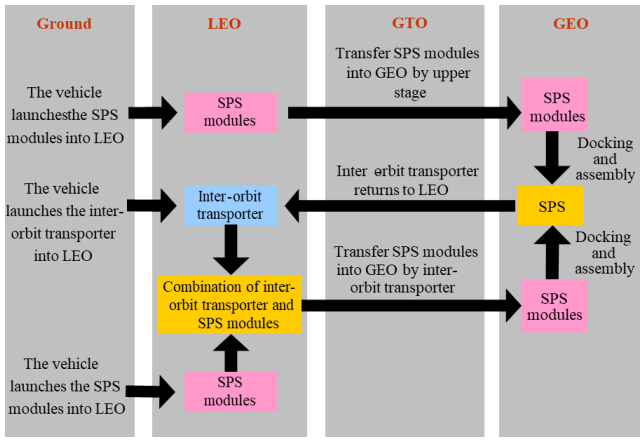


Fig. 14. Schematic Diagram of GEO Assembly and Transportation mode.

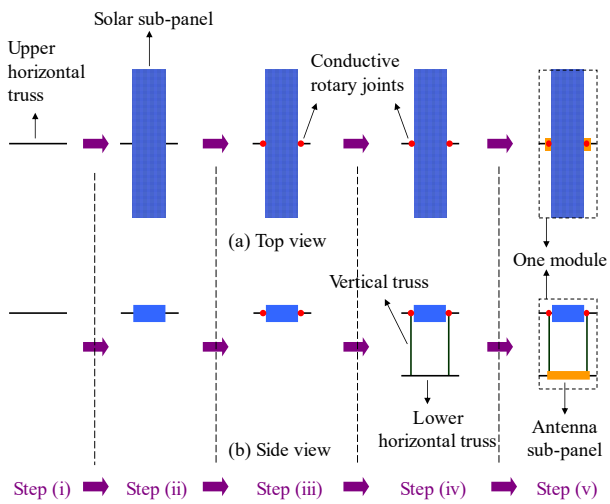


Fig. 15. The assembly sequence of an SPG-MPT module of the MMR-SPS.

The MMR-SPS comprises solar array modules, truss modules, antenna modules, and other modules designed for convenient assembly in space. The pivotal aspect of this updated concept is that it consists of 50 identical SPG-MPT modules, and all modules are mutually independent.

The assembly sequence of an individual module is depicted in Fig. 15. Firstly, the solar array modules are deployed and assembled in orbit, followed by the sequential assembly of the corresponding conductive rotary joints and upper south-north trusses. Secondly, two connecting perpendicular trusses are assembled to the south-north truss and deployed in the *z*-direction. Thirdly, the deployed lower south-north truss is assembled between the two connecting trusses. Fourthly, the antenna modules are deployed and assembled based on the lower south-north truss. Finally, all cables and electrical equipment are assembled.

Space assembly robots are employed to complete the assembly, providing the functions of grabbing assembly modules, transporting modules to specific locations, assembling modules, and assisting in the deployment of modules. The space assembly robot system mainly includes platform mobile robots and free maneuvering robots. Platform mobile robots move and complete space assembly and maintenance operations through a moving platform on a truss or other structures. The mechanical arm operation system of the International Space Station is a typical platform mobile robot [39]. Almost all large structures of the space station are assembled through the mechanical arm operation system. Free maneuvering robots have the ability to move unhindered along orbits. They are equipped with a mechanical arm system that can achieve rapid transportation and assembly of payloads, facilitating the efficient construction of ultra-large space structures. Currently, research in free maneuvering robots is still in its early stages. All the robots are launched to the orbit together with the assembly modules. After that, they operate independently. Considering the assembly efficiency, it is estimated that a minimum of several dozen robots must operate simultaneously.

Since one solar sub-panel corresponds to one antenna sub-panel, the power generated by the solar sub-panel can be transmitted to the antenna sub-panel through two cables installed on the perpendicular trusses. This approach offers significant advantages for on-orbit assembly. Once an individual SPG-MPT module is assembled, it can operate and transmit power immediately. Subsequently, all SPG-MPT modules are repeatedly assembled, and the construction of the entire SPS system involves continuous addition of modules in the *y* direction. Accidental damage such as debris impact will not affect the operation of the entire system, and each module can be maintained and repaired independently.

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V. LIFE-CYCLE COST ANALYSIS

The life-cycle cost analysis of an SSP system is conducted following the flow chart in Fig. 16. The entire SSP system comprises a space segment and a ground segment. The analysis process begins by inputting the basic system parameters and selecting and analyzing the relevant parameters for both the ground and space segments. Subsequently, the design cost and development cost are calculated, followed by the computation of transportation costs based on specified parameters. The construction cost is determined according to assembly parameters, and operation and maintenance costs are calculated based on related parameters. Estimations are made for disposition

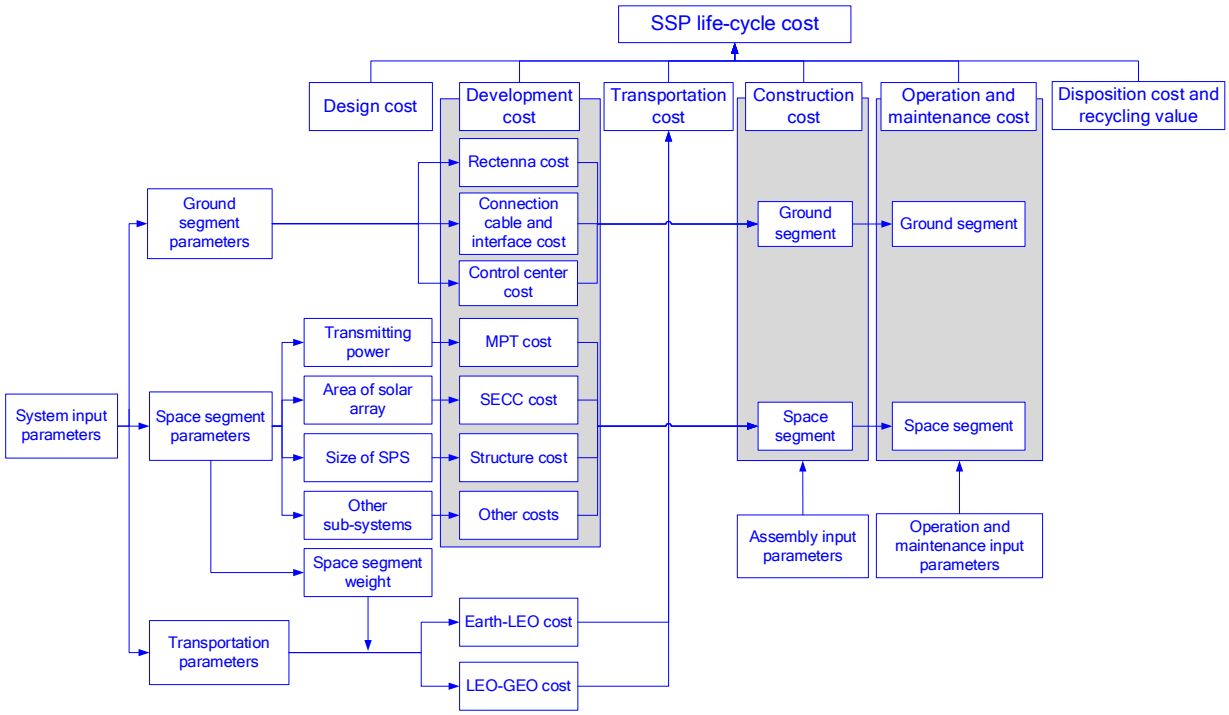


Fig. 16. The flow chart of life-cycle cost analysis of an SSP

TABLE II

The cost analysis result of the space segment (million US Dollars)

	Design	Development	Transportation ($1,000/kg)	Construction	Operation & maintenance	Disposition & recycling	Total
SECC[a] sub-system	7	2570	850	855	226	255	4763
PTM[b] sub-system	14	585	600	355	255	193	2002
MPT[c] sub-system	7	1800	2000	783	212	600	5402
Structure sub-system	14	410	770	712	42	–975	973
AOC[d] sub-system	14	455	40	70	770	8	1357
TM[e] sub-system	14	56	105	70	85	28	358
ISRM[f] sub-system	21	140	52	35	85	14	347
Total	91	6016	4417	2880	1675	123	15202

a SECC — solar energy collection and conversion.
b PTM — power transmission and management.
c MPT — microwave power transmission.
d AOC — structure, attitude and orbit control.
e TM — thermal management.
f ISRM — information and system operation management.

TABLE III

The cost analysis result of the ground segment (million US Dollars)

	Design	Development	Construction	Operation & Maintenance	Disposition & Recycling	Total
Rectenna	7	1420	282	300	-28	1981
Connect Cable and Interface	1.4	14	7	21	-2.8	40.6
Control Center	1.4	5.7	1.4	42	-1.4	49.1
Total	9.8	1439.7	290.4	363	-32.2	2070.7

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TABLE IV
Summary of a 1 GW MMR-SPS

System	Parameter	Value
SPS system	Orbit	GEO
	Supply power /GW	~1
	Total efficiency	~12.3%
	Total mass /t	~9160
	Life-cycle cost/M$	~17300
Solar Energy Collection and Conversion sub-system	Efficiency of PV cell	~40%
	Number of solar sub-panels	50
	Number of solar array modules	600
	Area of solar panel /km^2	~6
	Output power /GW	~2.4
	Voltage of solar array module /V	~500
	Mass/t	~1800
Microwave Power Transmission sub-system	Frequency /GHz	5.8
	Efficiency of WPT	~54%
	Size of transmission antenna /km	0.1×10.5
	Number of antenna sub-panels	50
	Size of antenna array modules /m	5
	Deform of antenna array module /mm	< 8
	Transmission power of an antenna sub-panel /MW	32
	Mass/t	~4200
Power Transmission and Management sub-system	Voltage of main bus /kV	5
	Number of rotary joints	100
	Mass/t	~1360
Other sub-system	Mass/t	1800

costs and recycling value. Ultimately, the sum of these costs represents the life-cycle cost of a space solar power system.

A cost analysis of the original MR-SPS concept using the above approach is provided in [24]. An update of the cost data for the proposed MMR-SPS is presented in Table II and Table III, showing the primary cost analysis results for the space segment and the ground segment, respectively. The calculated total cost of an MMR-SPS over a 30-year period is approximately 17.3 billion US dollars. Assuming a continuous output power of 1 GW, the total life-cycle energy supplied by a SPS is about 262.8 billion kWh. The cost per kWh is calculated to be approximately 6.6 US cents. Compared with the total cost estimation in [24], the cost estimation for MMR-SPS is lowered by more than 30%. The cost data also serves as the design requirement for future SPS development efforts and must be continuously amended as research and technology progress.

VI. CONCLUSION

In summary, this paper introduces and analyzes the primary scheme of an MMR-SPS concept, outlining the key parameters of a typical 1 GW MMR-SPS as shown in Table IV. The innovative distribution layouts of solar sub-panels and antenna sub-panels offer a solution to the challenges associated with long-distance, high-voltage electric power transmission in space. Each antenna sub-panel, paired with the corresponding solar sub-panel, constitutes an independent SPG-MPT module, leading to a modular SPS configuration. This modularity effectively reduces the complexity and weight of the space power transmission and management subsystem.

A distinctive feature of the proposed MMR-SPS concept is the elongated aperture of the transmitting antenna panel, posing challenges in high-precision MPT. To address this, the retro-reflective beamforming scheme is introduced. With a numerical calculation model, retro-reflective beamforming proves to be necessary and effective in compensating for the impact of antenna position deviation and aperture deformation.

The assembly of the entire SPS is also facilitated by the modular configuration, allowing each module to operate independently. Moreover, the modular nature enables each unit to transmit power immediately after assembly, improving the utilization efficiency of the entire SPS.

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